Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2343112	Advance Data Structure & Analysis	03	–	--	03	--	–	03

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II					
2343112	Advance Data Structure & Analysis	20	20	40	60	02	--	--	100

Course Objective: Students will able to learn:

Sr. No.	Course Objectives
1	To learn mathematical background for analysis of algorithm
2	To learn various advanced data structures.
3	To learn greed approach to solve problems.
4	To learn backtracking algorithm and maximum flow networks.
5	To learn dynamic programming methods.
6	To understand the concept of pattern matching.

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Course Outcomes:

On successful completion, of course, learner/student will be able to:

Sr.No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Understand methods for analysis of algorithms and solve recurrence problems.	L1, L2
2	Choose appropriate advanced data structures for a given problem and calculate its complexity.	L2, L3, L4
3	Analyze the greedy programming technique to solve problems.	L2, L3, L4, L5
4	Evaluate and analyze the backtracking algorithm and understand maximum flow networks.	L2, L3, L4, L5
5	Analyze the dynamic programming technique to solve problems.	L2, L3, L4, L5
6	Select a proper pattern matching algorithm for a given problem.	L3, L4, L5

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mappings
0	**Prerequisite**	Overview of Data Structures: Revision of basic data structures (arrays, stacks, queues, linked lists, trees).	**01**	
I	**Introduction to Analysis of Algorithms**	Fundamentals of the analysis of algorithms: Time and Space complexity, Asymptotic notation, Recurrence Relations: Methods to solve recurrence relations in algorithms (Substitution, Recursion tree, Master theorem). **Self-learning Topics:** Solve problems on analysis of algorithms.	**03**	**CO1**

II	**Advanced Data Structures**	Introduction. AVL trees, B tree, B tree operations, B+ tree, Red-Black Trees, tries data structures, time complexity analysis of all problems. Graphs, Representation, Graph Traversals: Breadth First Search, Depth First Search. **Self-learning Topics:** Solve problems on AVL trees, B tree, B+ tree etc.	**08**	**CO2**
III	**Greedy algorithms and Applications**	Introduction and properties of greedy algorithms, Fractional Knapsack problem, Minimum Spanning Trees (Prim's and Kruskal's algorithms), Job sequencing with deadlines, Optimal storage on tapes, Analysis of All problems. **Self-learning Topics:** Solve problems on Spanning Trees, Knapsack etc.	**06**	**CO3**
IV	**Backtracking and Maximum flow Networks**	Backtracking Techniques: Introduction, N-Queens problem, sum of subsets problem, graph coloring, Hamiltonian cycles. Introduction to flow networks, Augmenting Paths Residual Network, Ford Fulkerson method, Applications of Flow Networks in real-world problems. **Self-learning Topics:** Solve problems N-Queens, Hamiltonian cycles, Augmenting Paths Residual Network etc.	**07**	**CO4**
V	**Dynamic Algorithms**	Introduction Dynamic algorithms, Greedy vs. Dynamic algorithms, Single source shortest path- Dijkstra's Algorithm, Bellman Ford Algorithm, All pair shortest path- Floyd Warshall Algorithm, 0/1 knapsack problem, Travelling salesman problem, Analysis of All problems. **Self-learning Topics:** Solve problems on shortest path- Dijkstra's Algorithm etc.	**08**	**CO5**
VI	**String Matching Algorithms**	Introduction. Naïve string matching algorithm, Rabin-Karp algorithm, Knuth-Morris-Pratt(KMP) algorithm, Longest common subsequence(LCS), Analysis of All problems, **Applications:** Text searching, DNA sequencing, and data compression. **Self-learning Topics:** Solve problems on DNA sequencing, and data compression.	**06**	**CO6**
Note: No questions will be asked in the end-semester exam from self-study topics. However, students are encouraged to explore these topics for a better understanding of the subject.				

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Text Books and References:

Sr. No	Title	Authors	Publisher	Edition	Year
1	Introduction to Algorithms	Cormen, Leiserson, Rivest, Stein	PHI	3rd Edition	2011
2	Algorithm Design	Jon Kleinberg, Éva Tardos	Pearson	1st Edition	2006
3	Data Structures and Algorithm Analysis in C++	Mark Allen Weiss	Pearson	4th Edition	2013
4	Introduction to the Design and Analysis of Algorithms	Anany Levitin	Pearson	3rd Edition	2011
5	Algorithms	Robert Sedgewick, Kevin Wayne	Addison-Wesley	4th Edition	2011

Online Resources:

S. No.	Website Name	URL	Modules Covered
1	NPTEL	https://archive.nptel.ac.in/courses/106/106/106106200/	M1
2	NPTEL	https://archive.nptel.ac.in/courses/106/105/106105085/	M2
3	NPTEL	https://archive.nptel.ac.in/courses/106/104/106104120/	M3
4	Coursera	https://www.coursera.org/learn/algorithms-part1	M1-M3
5	MIT OpenCourseWare	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-006-introduction-to-algorithms-fall-2011/	M1-M6

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Assessment:

Internal Assessment Test (IAT) for 20 marks each:

- IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

End Semester Theory Examination:

➢ **Question paper format**

- Question Paper will comprise a total of **six questions each carrying 15 marks Q.1** will be **compulsory** and should **cover the maximum contents of the syllabus**

- **Remaining questions** will be **mixed in nature** (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)

- A total of **four questions** need to be answered.